

Jack Brewer

+44 7402 279724 | brewer.jack@protonmail.com | github.com/br-jack | linkedin.com/in/jack-brewer-4a93832b1

EDUCATION

University of Bristol

Computer Science MEng; Projected 2:1

Sep. 2023 — Jun. 2027

- Design Verification - functional verification using e language over HDL design, covering constrained random testing, functional coverage and assertion-based verification
- High Performance Computing - optimised fluid simulation across serial, SIMD, and multi-threaded (up to 4x144 core) implementations, analysing performance characteristics of the Isambard 3 supercomputer
- Computer Graphics - implemented a graphics engine in C++ and SDL from scratch, including 3D transformations, texture mapping, lighting and camera movement using linear algebra
- Software Engineering Project - developed a full-stack web app in a team with an external client, leading backend development and acting as client liaison across a nine month project

WORK EXPERIENCE

Teaching Assistant

Sep. 2025 — May 2026

University of Bristol

Bristol

- Mentored team of four students in weekly meetings for Software Engineering Project
- Acted as a critical point of contact for students to discuss project progress and challenges
- Helped run workshop for less confident students to become proficient with technologies such as Git

Student Developer - Airship Game | JavaScript

May 2025 — Sep. 2025

University of Bristol

Bristol

- Identified issues with a programming game used by hundreds of prospective pupils every year
- Went through large, outdated codebase with next to no documentation
- Updated user documentation to reflect current capabilities and make usage clearer

Student Helper

Sep. 2024 — Present

University of Bristol

Bristol

- Lead open day tours and answering questions from potential new students
- Helping teach prospective students in mock lab by answering programming questions and helping with bug-fixes

Campsite Warden

Aug. 2025 — Present (Seasonal)

Llyn Gwynant Campsite

Gwynedd

- Managed customer registration using complex site management and POS systems in high-pressure peak periods
- Maintained site facilities and operational standards consistently across extended shifts
- Prepared and served food and drinks

PROJECTS

Merlin - Learning App, TypeScript

Jan. 2026 — Present

- Advised on software architecture decisions for TypeScript learning app
- Containerised application using Docker

Hackathon (Brishack 2025 - Eco), Rust

Feb. 2025

- Made an app for a Library of Things to be used in local areas to allow people to borrow objects and tools
- Designed and helped implement API endpoints
- Designed and created script for SQL database

OTHER QUALIFICATIONS

Outdoor Pursuits 16 Hour First Aid Level 3 (VTQ), ProTrainings

Valid 28 Sep. 2024 - 27 Sep. 2027

SKILLS

- **Programming Languages** (Proficient): C#, TypeScript, C
- **Programming Languages** (Basic): e Language (Specman), JavaScript, Python, C++
- **Technologies**: React, NestJS, Git, AWS, SQLite, PostgreSQL